RAILWAY RESTAURANT

A marginal-cost business model for high-quality, low-cost food on Britain’s mainlines

PRICE × QUANTITY × PRODUCTIVITY

Business model and investment case

Prepared as an independent policy and commercial model for discussion with HM Treasury, the Department for Transport and mainline train operators.

5 November 2026 | Great Britain | GBP nominal prices

£8.40	350	£63k
Target average basket	Daily transactions	Annual operating surplus

Important status

This is an illustrative feasibility model, not an operator-approved business plan, an LSE publication, or a Treasury forecast. It uses published market references plus explicit planning assumptions. Actual train capacity, operator concession terms, route demand, fit-out costs, supplier invoices and tax treatment require diligence before investment.

Executive summary

The proposal is a high-quality, value-priced restaurant service carried on selected long-distance trains. It is designed around existing mainline catering space rather than purchasing new rolling stock: three compatible café or buffet vehicles are refurbished with modular food finishing equipment, upgraded presentation, cold storage, digital ordering and a small reserved dining area. A route-side commissary batch-prepares a short menu; meals are chilled, transferred in insulated food-safe logistics, regenerated onboard and delivered to reserved seats or served in the restaurant section.

Investment thesis

The opportunity is not to make food cheap by reducing quality. It is to lower the cost of delivering a consistent, attractive meal by standardising recipes, pooling procurement, using demand data before production, reducing waste, and keeping complex cooking off the train.

Base-case model at a glance

Metric	Base assumption / result
Equipped departures	8 selected one-way departures per day
Passenger market exposed to offer	3,000 boardings per day (8 x 375; planning assumption)
Food transactions	350 per day; 11.7% conversion
Average customer basket	£8.40 including VAT; £7.00 net revenue
Short-run marginal cost	£3.26 per additional transaction before roster and other step-fixed costs
Daily break-even volume	About 294 transactions
Annual operating days	300
Annual net sales	£735,000
Operating surplus after capital recovery	About £63,450 per year; 8.6% of net sales

Recommendation

- Run a 12-month commercial pilot on one high-volume intercity corridor using three existing compatible catering vehicles and one commissary.
- Price the core hot meal range at £6.95-£7.95, with a target average basket of £8.40 including a drink or side. Keep the offer transparent and below a premium-restaurant price point.
- Tender or negotiate the operator concession on an open-book basis, with a modelled 8% share of net sales and clear service, price and food-safety obligations.
- Scale only after daily transaction counts reliably exceed the 294-order break-even threshold and the pilot demonstrates at least 8% operating margin after a capital recovery allowance.

The model is deliberately conservative about labour: the 41-hour daily roster is treated as step-fixed within the current schedule, not as a negligible per-meal cost. The reported surplus is before interest and tax, but after a straight-line capital recovery charge.

1. Market context and evidence base

The addressable market is large in aggregate, but the business must win a small share of the passengers actually travelling on the selected departures. ORR reports 1.83 billion rail journeys in Great Britain in April 2025-March 2026, the highest annual total since 1920. In April-June 2026, 461 million journeys were made, 2% more than the same quarter a year earlier; passenger revenue was £3.2 billion in that quarter. These numbers establish rail as a large distribution channel, but they do not tell us what proportion of passengers buy food onboard. [1][2]

A live benchmark already exists. LNER operates Café Bar service, seat delivery through its “Let’s Eat At Your Seat” QR ordering service, locally sourced food and drink, and a published value offer: a £6.50 meal deal, a £7 cheeseburger-and-soft-drink deal and a £4.50 bakery-and-hot-drink deal. This validates the customer interface and shows that the proposed concept should upgrade and deepen existing onboard catering rather than assume the market is empty. [3]

There is an unusually important data gap. In a written parliamentary answer dated 8 September 2026, the Department for Transport said it does not centrally hold a complete, consistently defined breakdown of onboard catering revenue, cost and net profit for the listed train operators. Some operator or contractor data exists but is commercially sensitive. A pilot should therefore include open-book measurement as a core output, not just a food-service launch. [4]

Current cost and regulatory signals

Evidence point	Implication for the model
Food and non-alcoholic beverage inflation: 1.3% year-on-year in September 2026 (ONS). [5]	Ingredient inflation is not the sole obstacle; labour, logistics, service and rail access must be managed.
National Living Wage: £12.71/hour for workers aged 21+ from April 2026. [6]	Use a loaded labour cost, not the headline wage, in the unit economics.
Employer Class 1 NIC: 15% above the £96/week secondary threshold in 2026-27. [7]	Roster cost must include employer on-costs and paid non-productive time.
Catering and hot food are generally standard-rated for VAT. [8]	The base case treats the average basket as 20% VAT inclusive; confirm each product classification with HMRC.
WRAP estimates food waste costs hospitality and food service £3.2bn a year. [9]	Forecasting, short menus and batch sizes are economic controls, not just sustainability measures.

Evidence limitation

The 3,000 daily passenger boardings, 11.7% take-up, recipe mix, roster hours, concession share and £450,000 refit are model assumptions. They are not ORR or operator-reported facts. The DfT data gap makes a measured pilot especially valuable.

2. The operating model

A restaurant experience without a full restaurant kitchen on every train

The proposed service is an onboard restaurant brand integrated with an existing train operator's café or buffet function. Three compatible vehicles are fitted out at an assumed £150,000 each. Each has a compact finishing galley, refrigerated holding, temperature monitoring, a premium but durable counter, POS/QR capability, and a 24-32 seat dining zone where vehicle geometry and accessibility allow. Most customers can also order from their own seats.

Stage	Operating method	Marginal-cost mechanism
Demand capture	Bookings, journey load, QR pre-orders and prior sales inform production by departure.	Pre-orders reduce safety stock and unsold meals.
Central preparation	One route-side commissary produces a short menu in batch; recipes share ingredients and preparation steps.	Labour and equipment are spread over larger batches.
Cold-chain dispatch	Meals are portioned, labelled and loaded by vehicle and departure.	Fewer SKUs and standard containers reduce handling cost.
Onboard finishing	Regenerate chilled meals using validated equipment; do not run a full cooking line.	Lower galley complexity, training and energy per transaction.
Customer service	Pre-order to seat, counter collection and reserved dining seats. Contactless payment.	Shorter queue time and fewer transactions requiring bespoke service.
Data loop	Record sell-through, waste, refunds, dwell time, temperature logs and customer ratings.	Daily menu and production decisions respond to observed demand.

Illustrative service envelope

Planning variable	Assumption
Selected departures	8 one-way services per day
Average boardings per selected departure	375
Total exposed passenger boardings	3,000 per day
Target transactions	350 per day; approximately 44 per departure
Restaurant seating	Up to 32 seats per fitted vehicle; actual seats subject to safety and accessibility approval
Additional service	Seat delivery / collection to passengers outside the dining zone
Operating days	300 per year; allowance for timetable variation, holidays and planned maintenance

The service plan assumes about 154 dine-in covers per day (approximately 19 per departure at 60% use of 32 seats) and about 196 meals or baskets collected or delivered to other passengers. Those are mix assumptions, not separate demand forecasts. The 350 total covers remain the single financial volume measure.

Why three vehicles?

The three-vehicle assumption is intended to cover a set of train diagrams and turnarounds while equipping eight selected departures. It must be validated against the chosen operator's actual rolling-stock diagrams, maintenance windows, formation limits and depot arrangements. No new train is purchased in this model.

3. Menu architecture and price position

The menu is “Anglo-futurist” in the commercial sense: contemporary British produce, clean presentation, consistent portions, modern packaging and a restrained menu that feels designed rather than mass-produced. It is not an expensive fine-dining kitchen. The offer is premium in reliability and execution, not in unnecessary ingredient complexity.

Product	Menu price incl. VAT	Design logic
Yorkshire chicken, leek and mustard mash	£6.95	Popular protein, high-yield thigh meat, shared root vegetables and sauce base.
British beef, ale and pearl-barley pot	£7.50	Batch-braised filling; uses a stable stew / pie format and common vegetables.
Cod, pea and cheddar fish pie	£7.95	Premium anchor dish; cod loin portioned with potato, peas and white sauce.
Mushroom, barley and roast-root pot (vegan)	£6.95	Lowest-cost main; plant-forward but protein-containing pulses and grains.
Soup / seasonal side	£2.50	Use menu trim and seasonal surplus only where safe and planned.
Coffee / tea / soft drink	£2.25-£3.20	Drinks lift average basket value; hot drinks use standardised cups and recipes.
Average transaction in base case	£8.40	Weighted average across mains, drinks, sides and meal bundles; not one menu item.

Public price benchmark

LNER’s published onboard menu lists a £6.50 sandwich, crisps and drink meal deal and a £7 cheeseburger with soft drink. It also lists £4.50 bakery-and-hot-drink offer. The proposed £6.95-£7.95 hot main range is therefore positioned as a higher-quality hot-meal alternative, while the £8.40 average basket assumes an add-on drink or side. It should not be marketed as cheaper than every existing train product. [3]

Menu engineering rules

- Hold the regular menu to four mains, one soup/side, two desserts and a controlled drinks range. Rotate one seasonal dish rather than expanding the permanent SKU count.
- Share potatoes, carrots, onions, peas, dairy sauce, stock and grains across dishes; buy to recipe yield rather than to dish-by-dish guesswork.
- Keep one premium anchor (fish pie) but make the chicken and vegan dishes the volume leaders; use a weekly menu mix to protect average food cost.
- Publish allergens and ingredient substitutions before purchase and on delivery; maintain dish-specific production batches and traceability.

4. Recipe cost cards

The following are indicative ingredient costs per finished portion, based on a 450 g main meal. They use public UK wholesale reference prices checked in September-October 2026 where available; secondary ingredients without a matching public price line are explicit allowance estimates. The recipe costs are model inputs, not supplier quotations. A 5% trim / preparation loss is included in the rounded totals. [10]

Recipe (450 g)	Key raw-material cost build	Cost per portion
Yorkshire chicken, leek & mustard mash	150 g chicken thigh (£0.59); 200 g potato (£0.14); carrot/onion/leek; cream, milk, flour, butter, rapeseed oil, stock and mustard.	£1.13
Beef, ale & pearl-barley pot	120 g beef mince (£0.90); 50 g pearl barley; 180 g potato (£0.12); carrot/onion/tomato; ale, stock, butter and herbs.	£1.52
Cod, pea & cheddar fish pie	120 g cod pro-rated from £3.30 per 180-200 g loin; potato, peas, milk, flour, butter, cheddar and herbs.	£2.72
Mushroom, barley & root pot	Mushroom, pearl barley, chickpeas, potato, carrots, onion, rapeseed oil, stock and herbs. Mushroom / pulse prices are planning allowances.	£1.00

Published supplier references used

Ingredient	Reference price	Costing use
Boneless skinless chicken thigh	£3.90/kg	150 g = £0.59
Beef mince	£7.50/kg	120 g = £0.90
Cod loin	£3.30 each (180-200 g)	120 g pro-rata ≈ £2.08
Potatoes	£0.68/kg	180 g = £0.12
Carrots / onions	£0.75/kg each	50 g = £0.04 per ingredient
Chopped tomatoes	£1.13/kg	50 g = £0.06
Frozen peas	£2.79/kg	50 g = £0.14
Milk / double cream	£0.85/litre / £3.79/litre	50 ml = £0.04 / £0.19

Price provenance: ChefSums compiled public JJ Foodservice listing prices checked on specified dates in September and October 2026. It expressly states these are snapshots from one supplier, not market averages; invoices, pack sizes, delivery and negotiated contracts will differ. The cod pro-rata calculation uses the listed £3.30 loin price. [10]

The blended £1.85 per transaction food-and-drink material allowance used in the business model is higher than the weighted main-recipe ingredient cost. It includes the assumed menu mix, beverage inputs and minor add-ons. It should be replaced with a live recipe and sales-mix model once the pilot is trading.

5. Cost inputs, payroll and tax

Packaging

JJ Foodservice lists a 650 ml heavy-duty microwave container with lid at approximately £0.13 each on collection at a 20+ quantity tier and approximately £0.16 each on delivery. The model uses £0.16 for the tray and lid, then adds an allowance for napkin, cutlery, label, bag, cup allocation and incidental packaging to reach £0.28 per customer transaction. A reusable dine-in plate can reduce disposable packaging where the train's wash-up and storage arrangements support it. [11]

Labour

The statutory National Living Wage for workers aged 21+ is £12.71 per hour from 1 April 2026. Employer Class 1 National Insurance is 15% above the £96 per week secondary threshold for 2026-27. The model uses a fully loaded £17.50 per paid hour to allow for employer NI, paid leave, pension, training, supervision and roster inefficiency. This is a commercial planning rate, not the legal minimum or a claim about the operator's actual wage bill. [6][7]

Daily roster component	Paid hours	Cost at £17.50/hour
Onboard service / host cover across selected departures	25	£437.50
Commissary preparation, packing and dispatch	16	£280.00
Total daily paid hours	41	£717.50

VAT and food safety

- The financial model assumes the service is catering and standard-rated at 20% VAT. HMRC guidance distinguishes ordinary catering and hot takeaway from most retail food, which is often zero-rated. All modelled meal baskets are treated as standard-rated to avoid relying on a favourable product-by-product classification. Obtain written VAT advice for the final menu, dine-in / takeaway arrangements and bundled offers. [8]
- A food business must register with the relevant local authority; the FSA says registration applies to food preparation, storage, handling, sale and distribution, including mobile or temporary operations. Food businesses must manage allergens, traceability, food safety procedures and recall processes. [12][13]
- FSA guidance gives 70°C for two minutes as a standard safe cooking combination; hot food held for service should be at or above 63°C, with a limited two-hour exception below that threshold. Reheating is limited to once. The cook-chill and onboard regeneration plan needs validated HACCP controls and logged temperatures. [14]

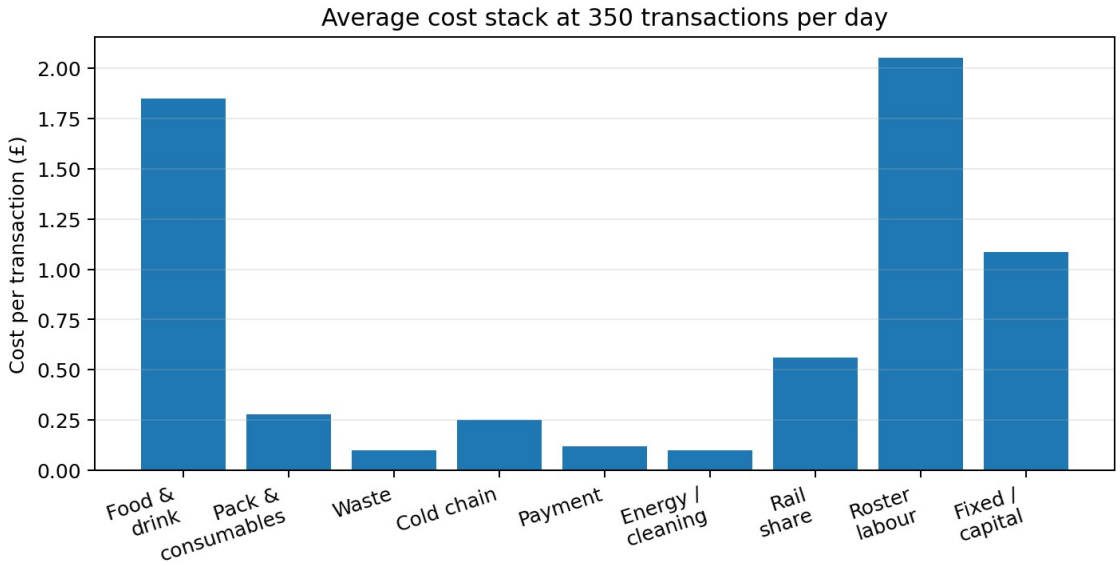
6. Unit economics: marginal cost and average cost

A distinction is essential. The true short-run marginal cost of one additional transaction is not the same as the fully allocated cost per meal. In the current roster, wages, vehicle fit-out recovery and management overhead are step-fixed for the day. Until the existing team or service capacity is exhausted, the next meal adds ingredients, packaging, waste, logistics, card cost, cleaning/energy and a revenue-linked operator concession fee. When another shift, vehicle or dispatch run is needed, marginal cost jumps.

Per transaction at £8.40 gross basket	£ / transaction	Basis
Net sales after assumed 20% VAT	7.00	£8.40 ÷ 1.20
Food and beverage ingredients	1.85	Weighted recipe and beverage mix
Packaging / cutlery / labels	0.28	Supplier tray reference plus accessory allowance
Food waste / trim allowance	0.10	Approx. 5% of ingredient inputs
Cold-chain transfer / handling	0.25	Planning allowance per order
Payment / ordering fees	0.12	Planning allowance
Energy and cleaning consumables	0.10	Planning allowance
Rail operator concession share	0.56	8% of £7.00 net sales
Short-run marginal cost	3.26	Sum of variable cost lines
Marginal contribution before roster / fixed costs	3.74	£7.00 - £3.26

Daily fixed and step-fixed cost

Cost item	£ per operating day	Calculation
Staff roster	£717.50	41 paid hours x £17.50 loaded rate
Other fixed operating cost and capital recovery	£380.00	£187.50 capital recovery + £192.50 site, systems, insurance, maintenance and management
Total fixed / step-fixed cost	£1,097.50	Applied before tax and finance



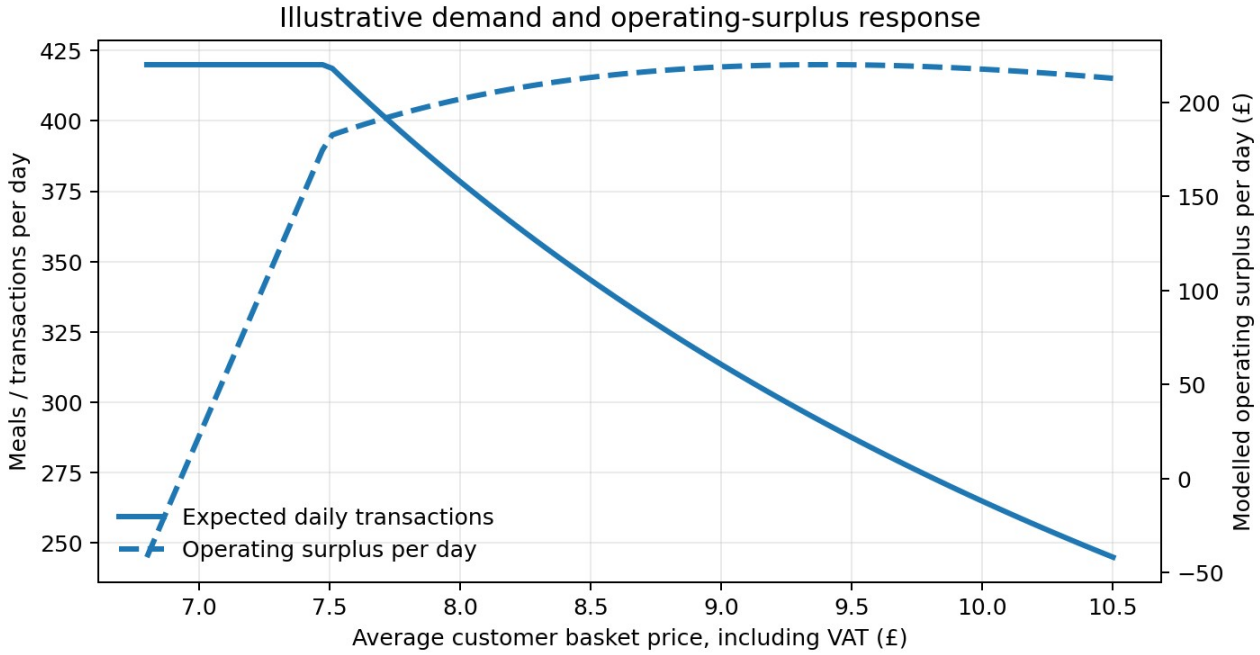
Average total operating cost at 350 transactions/day = £3.26 marginal cost + (£1,097.50 fixed/step-fixed daily cost ÷ 350) = approximately £6.40 per transaction, excluding VAT. Net revenue is £7.00; the base-case operating surplus is therefore about £0.60 per transaction.

7. Demand, price and equilibrium

The demand model is an explicit planning curve, not an observed rail-food elasticity study. It is calibrated to 350 transactions/day at an £8.40 average gross basket and uses a constant price elasticity of -1.6. It is capped at 420 transactions/day, the assumed practical sales capacity of the eight selected departures. The model allows the business to compare a lower value price with a higher price that earns more contribution per sale but reduces take-up.

Illustrative demand function

Q(P) = min[420, 350 x (P / £8.40)^(-1.6)]
P is the average customer basket price including VAT; Q is daily transactions. The -1.6 elasticity and the 420-transaction capacity are assumptions to be tested in the pilot.

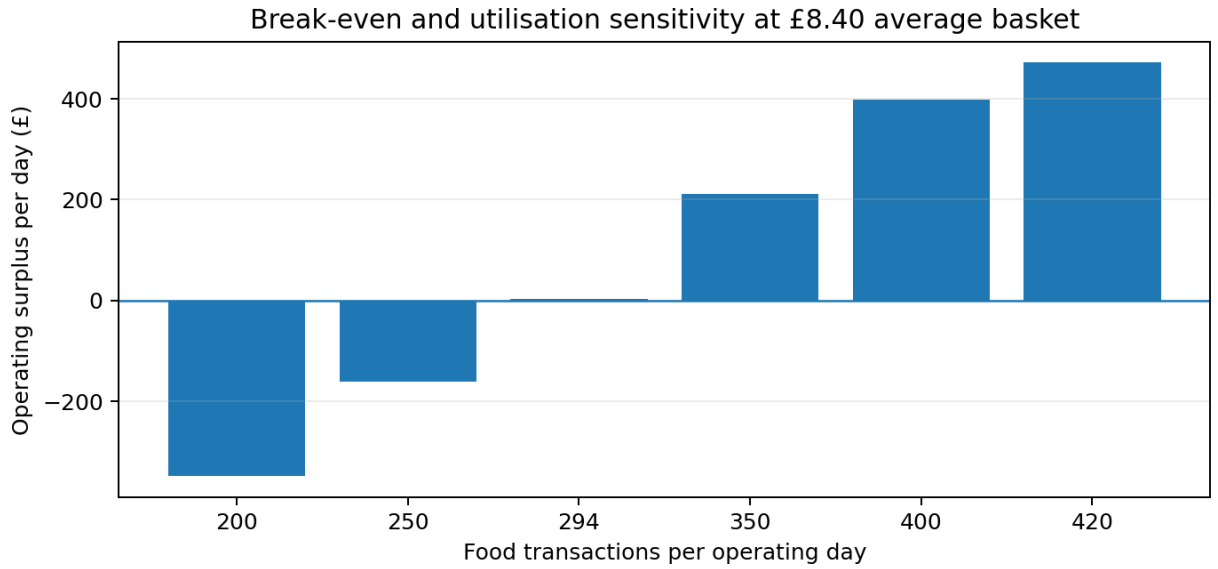


Basket price incl. VAT	Daily volume	Daily operating surplus	Interpretation
£7.50	420 (capacity-capped)	£182	Value price; high volume; constrained by service capacity.
£8.40	350	£212	Recommended accessible base case; 11.7% take-up.
£9.39	293	£220	Modelled profit-maximising price under assumed elasticity.
£10.00	265	£218	Higher price, lower volume; little extra surplus over £8.40.

The model's unconstrained profit-maximising point is approximately £9.39 gross and 293 transactions per day. The recommended £8.40 basket instead delivers about 350 transactions and roughly 96% of the modelled maximum daily surplus. That is an attractive trade: a roughly 10.5% lower average customer price and around 19% more transactions for only about £8.50 less daily surplus. The price should ultimately be selected using observed demand and customer outcomes, not this assumed elasticity alone.

8. Break-even, utilisation and profit margin

At the £8.40 gross basket, net sales are £7.00 and short-run marginal cost is £3.26, leaving £3.74 contribution per transaction. Total fixed and step-fixed cost is £1,097.50/day. Break-even is therefore approximately 294 transactions per day. This is the principal commercial hurdle: an attractive food margin on individual dishes does not rescue a service that fails to achieve enough daily purchases to cover its roster and rail-specific fixed costs.



Transactions/day	Daily operating surplus	Annual surplus (300 days)	Reading
200	-£350	-£104,850	Too little demand; restructure or do not launch.
250	-£163	-£48,750	Below break-even; price alone may not fix weak take-up.
294	Approximately £0	Approximately £1,000	Break-even threshold.
350	£212	£63,450	Base case; about 8.6% operating margin on net sales.
400	£398	£119,550	Good utilisation; requires service capacity to hold.
420	£472	£141,390	Modelled sales-capacity ceiling; verify staffing and galley throughput.

Base-case annual income statement

Line	Annual amount
Gross customer receipts incl. VAT	£882,000
VAT element (assumed 20% of gross)	£147,000
Net sales	£735,000
Variable costs incl. 8% operator sales share	£342,300
Staff roster cost	£215,250
Other fixed costs, including £56,250 annual capital recovery	£114,000
Modelled operating surplus before finance and tax	£63,450
Operating surplus / net sales	8.6%

This is a single service-unit planning case, not audited accounts. The annual surplus is after an £56,250 straight-line annual recovery allowance on £450,000 of refit capital; it is before interest and tax. Working capital is not expensed. The model assumes 300 trading days and no subsidy.

9. Capital requirement and sensitivity

Indicative launch funding

Use of funds	Amount	Comment
Refit and onboard equipment across three existing vehicles	£450,000	£150,000 per vehicle planning allowance; includes modular galley, chilled holding, finishing equipment, interior works, digital ordering/POS and engineering contingency.
Pre-opening, route integration and compliance	£50,000	Design, testing, menu validation, training, safety case support, legal and launch.
Opening working capital	£75,000	Food stock, packaging, payroll timing and supplier settlement buffer.
Total indicative funding envelope	£575,000	No purchase of train vehicles or infrastructure included.

The £450,000 fit-out is a planning assumption, not a quote from a train manufacturer, rolling-stock owner or operator. The three-vehicle fit-out must be validated against the actual vehicle type and operator's engineering requirements. If existing café vehicles can be used with lighter modular changes, the investment could be lower; if a full bespoke restaurant carriage is required, it could be much higher.

Sensitivity at the 350-order base case

Change from base	Approx. annual effect	Management response
Food and beverage input costs +20%	Operating surplus falls by about £38,850	Dual-source key ingredients; engineer alternative portions and seasonal substitutions.
Loaded labour rate +10%	Operating surplus falls by about £21,525	Simplify service, schedule against bookings, avoid overstaffing and protect pay.
Volume falls 15% (350 to 298)	Surplus falls to about £5,000/year	Trigger service redesign if the rolling 8-week average drops below break-even.
Volume rises to 400/day	Surplus rises to about £119,550/year	Expand only if safety, service time and replenishment capacity stay within limits.
Operator commission increases from 8% to 12% of net sales	Surplus falls by about £25,200/year	Negotiate a cap or lower rate in return for transparent performance data.

Investment hurdle

At base-case earnings, the £450,000 fit-out capital is recovered through the annual capital charge over eight years, but simple cash payback against the full £575,000 launch funding is about nine years if the base-case surplus is achieved and working capital is not released. This is not a high-return proposition at 350 orders/day. It becomes materially more attractive at 400-420 orders/day, or with a lower refit cost, shared commissary overhead, or operator-funded vehicle works. A pilot should not be scaled on the strength of menu gross margin alone.

10. Pilot, governance and policy design

12-month pilot gates

Phase	Months	Deliverable / gate
Commercial and engineering feasibility	0-2	Secure operator letter of intent; validate vehicle diagrams, access terms, seat count, galley power, water, waste and maintenance constraints; obtain fit-out quotations.
Recipe and supply-chain validation	1-3	Run supplier tenders; test 4 mains and 2 rotating sides; approve allergen matrix, shelf life, cook-chill validation and yields.
Fit-out and systems	3-6	Refit three existing compatible vehicles; integrate ticketing/load data, QR/POS, stock and temperature records; train staff.
Controlled launch	7-9	Start on selected departures; monitor take-up, service time, stock-outs, waste, refund rate, customer scores and contribution per order.
Scale / redesign decision	10-12	Scale only after eight consecutive weeks above 294 daily transactions, positive operating surplus, no material safety issues and verified customer price/value performance.

Treasury and DfT actions

- Enable a competitive, time-limited pilot concession on a high-volume intercity route; do not prescribe a route before operator capacity and passenger-flow data are supplied.
- Make any public support conditional on open-book reporting of sales, net of VAT, food input cost, paid hours, waste, concession payments and service availability.
- If capital support is considered, cap it, match it to private investment, pay against engineering and price milestones, and include clawback if the operator withdraws the value offer or fails to report outcomes.
- Require the operator to report the share of savings passed to passengers through menu prices, meal bundles or service frequency. Avoid permanent operating subsidy unless a distinct public-service case is demonstrated.
- Publish anonymised aggregate pilot data to close the current information gap about the economics of onboard catering. [4]

Core key performance indicators

KPI	Target / reporting rule
Transactions	At least 294/day break-even; 350/day base; report by departure and time band.
Average transaction value	£8.40 gross target; show main-only and add-on pricing separately.
Food-and-drink cost	Target £1.85/order blended; weekly recipe variance against invoice prices.
Waste	Target <=5% of edible inputs by cost at launch; trend toward <=3% after menu stabilisation.
Labour productivity	41 paid hours/day for 350 transactions; track meals per paid hour and queue time.
Availability / safety	Report stock-outs, temperature compliance, allergen incidents, recalls and refunds.
Price pass-through	Compare menu prices with the agreed price cap / basket index and published train benchmarks.

11. Risk register and conclusion

Risk	Why it matters	Control / mitigation
Passenger conversion below break-even	Fixed roster and vehicle costs remain even if demand is weak.	Pre-orders, timetable-specific forecasts, launch only on strongest trains, eight-week trigger for redesign.
Rolling-stock / operator approval	No independent right to install a restaurant service on a train.	Operator concession, rolling-stock owner approval, engineering and safety review before fit-out.
Food safety and cold-chain failure	Illness, legal exposure, waste and service suspension.	Registered food business, documented food-safety management, traceability, allergen controls and temperature logs.
Labour productivity overestimated	One extra employee or a longer service shift can erase the base margin.	Time-and-motion trial; staff-hours per departure; flexible roster bands; protect breaks and safe workload.
Ingredient price / availability shock	Fish, dairy, meat and packaging costs can be volatile.	Short menu, dual sourcing, portion control, seasonal substitutions, price re-openers.
Poor service / delays / stock-outs	Rail disruption changes demand and delivery windows.	Buffer stock for top lines, robust cold chain, QR ordering cut-off, refund protocol and disruption playbook.
VAT or contract misclassification	Could change net revenue or operator payment economics.	Tax review of each menu item and bundle; calculate concessions on clearly defined net-of-VAT sales.

Conclusion

A high-quality, low-cost railway restaurant is commercially plausible as a carefully engineered concession, not as a conventional restaurant transplanted onto a train. The model's economics are dominated by a small number of variables: daily transaction volume, loaded labour hours, the operator's commercial share, cost of the refit and the ability to use existing vehicles and commissary infrastructure.

Under the stated assumptions, the service breaks even at about 294 transactions per day and generates an operating surplus of around £63,450 a year at 350 transactions per day. The recommended £8.40 gross average basket is designed to maximise customer value while retaining most of the modelled profit-maximising surplus. This conclusion is conditional: the demand elasticity, transaction mix and fit-out budget are not yet proven.

Decision requested

Authorise a route-specific feasibility and tender phase, not immediate national rollout. Require three firm engineering / fit-out quotations, an operator concession term sheet, a passenger demand test and a 12-month open-book pilot. Scale only if the pilot proves both affordable menu prices and a sustainable return after all rail-specific costs.

Appendix A. Model formulae and definitions

Term	Definition used in this report
Gross basket price (P)	Customer spend including assumed 20% VAT.
Net revenue per transaction	P ÷ 1.20.
Variable non-commission cost	£2.70 per transaction: £1.85 food/drink + £0.28 packaging + £0.10 waste + £0.25 cold-chain/handling + £0.12 payment + £0.10 energy/cleaning.
Rail concession payment	8% of net sales; at £7.00 net revenue this is £0.56/order.
Short-run marginal cost	£2.70 + 8% of net revenue; £3.26 at the £8.40 gross basket.
Contribution per transaction	Net revenue less short-run marginal cost; £3.74 at base case.
Daily fixed / step-fixed costs	£717.50 roster + £380 other fixed costs = £1,097.50/day.
Break-even quantity	Daily fixed / step-fixed costs ÷ contribution = £1,097.50 ÷ £3.74 = 293.45, rounded up to 294.
Daily operating surplus	Q x contribution per transaction - £1,097.50.
Annual net sales	Q x net revenue x 300 operating days.
Demand curve	Q(P) = min[420, 350 x (P / £8.40)^(-1.6)].

How to interpret marginal cost

For the existing roster and equipment, marginal cost is approximately £3.26 at the base price. That does not mean the business can add unlimited orders for £3.26 each. Once kitchen capacity, onboard service hours, chilled storage, galley regeneration or train departure slots are exhausted, the next block of sales may require another worker, dispatch run, vehicle or shift. The cost curve is therefore stepwise, not perfectly flat. The 420-transaction ceiling is a planning constraint intended to prevent the model from extrapolating low marginal cost indefinitely.

Price equilibrium method

The model uses a constant elasticity demand curve with elasticity -1.6 and a price-linked concession fee of 8% of net sales. The gross price is divided by 1.20 to estimate net revenue. At each candidate price, expected quantity is estimated from the demand curve, short-run variable cost is calculated, and fixed daily costs are deducted. The price around £9.39 yields the highest daily surplus within the tested range. The lower £8.40 value price is recommended because it sells more meals while retaining approximately 96% of the modelled maximum surplus. This is a scenario calculation, not an empirical estimate of rail passenger willingness to pay.

Accounting boundary

The report treats the £450,000 refit as an asset recovered straight-line over eight years and 300 operating days per year, equivalent to £187.50 per operating day. The remaining £192.50/day of fixed costs is an aggregate planning allowance for route-side premises, systems, insurance, maintenance, management and vehicle-related support. Interest, corporation tax, financing fees, central group costs beyond this allowance, and any rolling-stock purchase are excluded. All numbers should be replaced by quotations and operator-specific contract data before a funding decision.

Appendix B. Sources and verification notes

Sources were checked on 5 November 2026. Public prices can change without notice. The listed supplier prices are snapshot references, not a tendered contract. Where a figure is an internal model assumption rather than a published statistic, this report labels it as such.

[1] Office of Rail and Road, “ORR says most journeys made on Great Britain’s rail network since 1920”, 18 June 2026. https://www.orr.gov.uk/search-news/orr-says-most-journeys-made-great-britains-rail-network-1920

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Research integrity statement

This report combines official rail, inflation, wage, tax and food-safety sources with public supplier price snapshots and an illustrative operating model. It is not an official LSE study, HM Treasury analysis, DfT position or train operator proposal. It should be read as a structured feasibility case for testing, not as a forecast or investment recommendation.